RAID LEVELS

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What is RAID?

- RAID (Redundant Arrays of Independent Disks) is a technique that makes use of a combination of multiple disks for storing the data instead of using a single disk for increased performance, data redundancy, or to protect data in the case of a drive failure.
- RAID (Redundant Array of Independent Disks) is like having backup copies of your important files stored in different places on several hard drives or solid-state drives (SSDs). If one drive stops working, your data is still safe because you have other copies stored on the other drives. It’s like having a safety net to protect your files from being lost if one of your drives breaks down.

RAID (Redundant Array of Independent Disks)

- RAID (Redundant Array of Independent Disks) in a Database Management System (DBMS) is a technology that combines multiple physical disk drives into a single logical unit for data storage.
- The main purpose of RAID is to improve data reliability, availability, and performance.
- There are different levels of RAID, each offering a balance of these benefits.

What is the problem with single disk storage?

- **No backup Disk**: If a single disk failure occurs, the whole system fails to perform. Due to data Redundancy if copy of data is stored in multiple disks then even if one disk failure occurs the system can still fetch data from the other redundant disk.
- **Performance:** If large amount of data is stored in a single disk it can degrade the performance and effectiveness of the system. This can be solved by using multiple disks with Redundant data.
- In RAID technique, the combined disks are considered as a single logical disk by the operating system. These individual disk uses different methods to store data. It depends on the type of RAID levels used.

RAID Levels in DBMS

- The different RAID levels used in DBMS are :
- **RAID 0**
- **RAID 1**
- **RAID 2**
- **RAID 3**
- **RAID 4**
- **RAID 5**
- **RAID 6**

RAID 0

- RAID 0 implements **data striping.**
- The data blocks are placed in multiple disks without redundancy.
- None of the disks are used for data redundancy so if one disk fails then all the data in the array is lost.
- In the figure, blocks “0,1,2,3” form a stripe.
- Instead of placing just one block into a disk at a time, we can work with two (or more) blocks placed into a disk before moving on to the next one.

RAID 0

Disk 0	Disk 1	Disk 2	Disk 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Evaluation

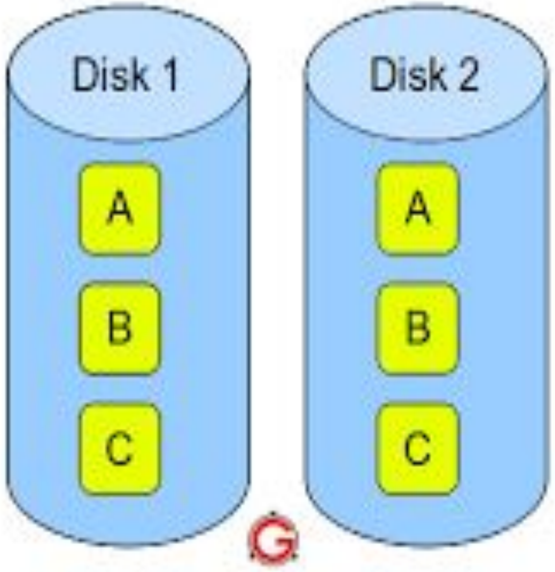
Reliability: 0
There is no duplication of data. Hence, a block once lost cannot be recovered.
Capacity: N*B
The entire space is being used to store data. Since there is no duplication, N disks each having B blocks are fully utilized.

Advantages &Disadvantages Of RAID 0

- **Advantages**
- It is easy to implement.
- It utilizes the storage capacity in a better way.
- All the disk space is utilized and hence performance is increased.
- Data requests can be on multiple disks and not on a single disk hence improving the throughput.
- **Disadvantages**
- Failure of one disk can lead to complete data loss in the respective array.
- No data Redundancy is implemented so one disk failure can lead to system failure.

RAID 1(mirroring)

- RAID 1 implements **mirroring** which means the data of one disk is replicated in another disk.
- This helps in preventing system failure as if one disk fails then the redundant disk takes over.
- More than one copy of each block is stored in a separate disk. Thus, every block has two (or more) copies, lying on different disks.
- The above figure shows a RAID-1 system mirroring level 2.
- RAID 0 was unable to tolerate any
- disk failure. But RAID 1 is capable of reliability.



RAID 1 – Blocks Mirrored. No Stripe. No parity.

RAID 1

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

Evaluation

- Assume a RAID system with mirroring level 2.
- **Reliability:** 1 to N/2
  1 disk failure can be handled for certain because blocks of that disk would have duplicates on some other disk. If we are lucky enough and disks 0 and 2 fail, then again this can be handled as the blocks of these disks have duplicates on disks 1 and 3. So, in the best case, N/2 disk failures can be handled.
- **Capacity**: N*B/2
  Only half the space is being used to store data. The other half is just a mirror of the already stored data.
- **Advantages**
- It covers complete redundancy.
- It can increase data security and speed.
- **Disadvantages**
- It is highly expensive.
- Storage capacity is less.

RAID-2 (Bit-Level Stripping with Dedicated Parity)

- In Raid-2, the error of the data is checked at every bit level. Here,
- we use Hamming Code Parity Method to find the error in the data.
- Two disks are used in this technique. One is used to store bit of each word in the disk and another is used to store error code correction (Parity bits) of data words. The structure of this RAID is complex, so it is not used commonly.

DISK 0	DISK 1	DISK 2	DISK 3	DISK 4	DISK 5
10	11	12	P(10)	P(11)	P(12)
14	15	16	P(14)	P(14)	P(16)
18	19	20	P(18)	P(18)	P(20)
22	23	24	P(22)	P(22)	P(24)

Advantages

- In case of Error Correction, it uses hamming code.
- It Uses one designated drive to store parity.

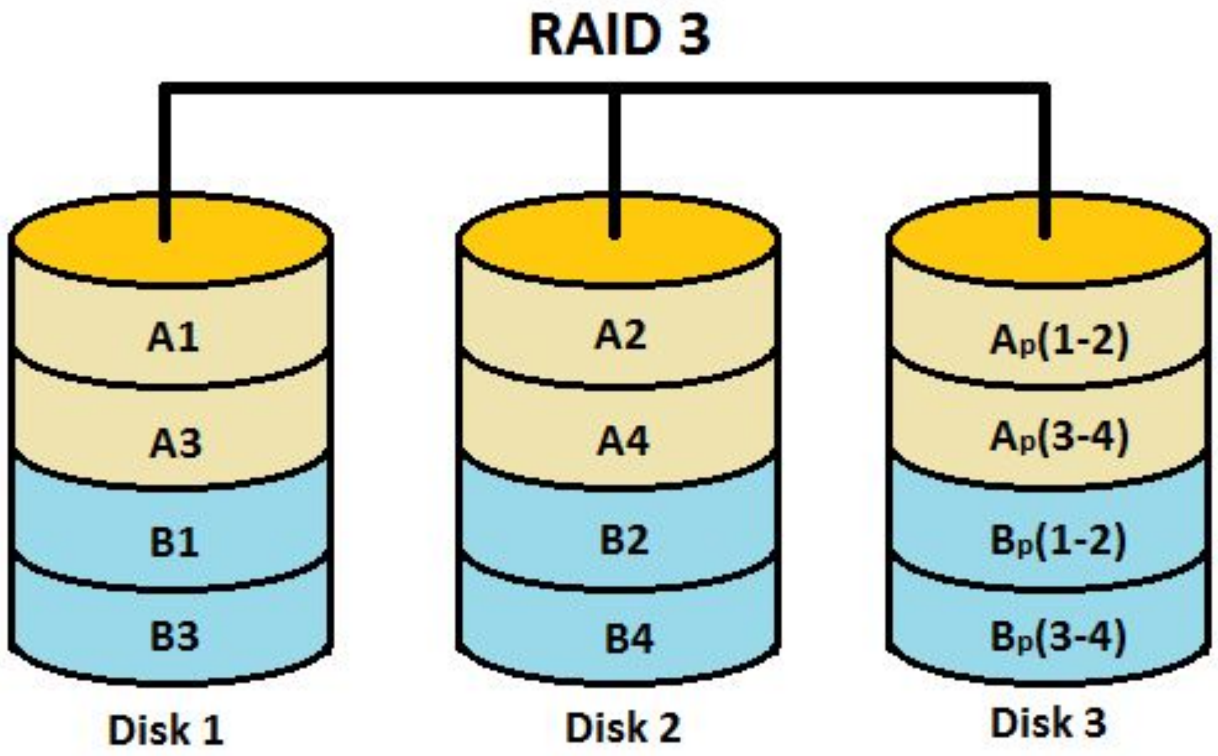
Disadvantages

- It has a complex structure and high cost due to extra drive.
- It requires an extra drive for error detection.

Here Disk 3, Disk 4 and Disk 5 stores the parity bits of Data stored in Disk 0, Disk 1, and Disk 2 respectively. Parity bits are used to detect the error in data.

RAID 3

- RAID 3 implements **byte-level striping of Data**. Data is stored across disks with their parity bits in a separate disk. The parity bits helps to reconstruct the data when there is a data loss.
- Whenever failure of the drive occurs, it helps in accessing the parity drive, through which we can reconstruct the data.
- Here Disk 3 contains the Parity bits for
- Disk 0, Disk 1. If data loss occurs, we
- can construct it with Disk 3.



Advantages & Disadvantages of RAID-3

- **Advantages**
- Data can be transferred in bulk.
- Data can be accessed in parallel.
- Data can be recovered with the help of parity bits.
- **Disadvantages**
- It requires an additional drive for parity.
- In the case of small-size files, it performs slowly.

RAID-4 (Block-Level Stripping with Dedicated Parity)

- RAID 4 implements **block-level striping** of data with dedicated parity drive. If only one of the data is lost in any disk then it can be reconstructed with the help of parity drive.
- Parity is calculated with the help of XOR operation over each data disk block.
- Instead of duplicating data, this adopts a parity-based approach.

RAID 4

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3

- In the figure, we can observe one column (disk) dedicated to parity.
- Parity is calculated using a simple XOR function. If the data bits are 0,0,0,1 the parity bit is XOR(0,0,0,1) = 1.
- If the data bits are 0,1,1,0 the parity bit is XOR(0,1,1,0) = 0. A simple approach is that an even number of ones results in parity 0, and an odd number of ones results in parity 1.

C1	C2	C3	C4	Parity
0	0	0	1	1
0	1	1	0	0

Assume that in the above figure, C3 is lost due to some disk failure. Then, we can recompute the data bit stored in C3 by looking at the values of all the other columns and the parity bit. This allows us to recover lost data.

RAID-4

- **Evaluation**
- **Reliability:** 1
  RAID-4 allows recovery of at most 1 disk failure (because of the way parity works). If more than one disk fails, there is no way to recover the data.
- **Capacity:** (N-1)*B
  One disk in the system is reserved for storing the parity. Hence, (N-1) disks are made available for data storage, each disk having B blocks.
- **Advantages**
- It helps in reconstructing the data if at most one data is lost.
- **Disadvantages**
- It can’t help reconstructing data when more than one is lost.

RAID-5 (Block-Level Stripping with Distributed Parity)

- This is a slight modification of the RAID-4 system where the only difference is that Evaluation
- Reliability: 1
- RAID-5 allows recovery of at most 1 disk failure (because of the way parity works). If more than one disk fails, there is no way to recover the data. This is identical to RAID-4.
- Capacity: (N-1)*B
- Overall, space equivalent to one disk is utilized in storing the parity. Hence, (N-1) disks are made available for data storage, each disk having B blocks.
- Advantages
- Data can be reconstructed using parity bits.
- It makes the performance better.
- Disadvantages
- Its technology is complex and extra space is required.
- If both discs get damaged, data will be lost forever.
- the parity rotates among the drives.

RAID 5

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
5	6	7	P1	4
10	11	P2	8	9
15	P3	12	13	14
P4	16	17	18	19

RAID-6 (Block-Level Stripping with two Parity Bits)

- Raid-6 helps when there is more than one disk failure. A pair of independent parities are generated and stored on multiple disks at this level. Ideally, you need four disk drives for this level.
- There are also hybrid RAIDs, which make use of more than one RAID level nested one after the other, to fulfill specific requirements.
- **Advantages**
- Very high data Accessibility.
- Fast read data transactions.
- **Disadvantages**
- Due to double parity, it has slow write data transactions.
- Extra space is required.

RAID 6

Disk 1	Disk 2	Disk 3	Disk 4
A1	B1	P(B1)	P(B1)
A2	P(B2)	P(B2)	B2
P(B3)	P(B3)	A3	B3
P(B4)	A4	A4	P(B4)

Advantages of RAID

- **Data redundancy:** By keeping numerous copies of the data on many disks, RAID can shield data from disk failures.
- **Performance enhancement:** RAID can enhance performance by distributing data over several drives, enabling the simultaneous execution of several read/write operations.
- **Scalability:** RAID is scalable, therefore by adding more disks to the array, the storage capacity may be expanded.
- **Versatility:** RAID is applicable to a wide range of devices, such as workstations, servers, and personal PCs

Disadvantages of RAID

- **Cost:** RAID implementation can be costly, particularly for arrays with large capacities.
- **Complexity:** The setup and management of RAID might be challenging.
- **Decreased performance:** The parity calculations necessary for some RAID configurations, including RAID 5 and RAID 6, may result in a decrease in speed.
- **Single point of failure:** RAID is not a comprehensive backup solution while offering data redundancy. The array's whole contents could be lost if the RAID controller malfunctions.